Trustworthy Collective Reasoning and Evaluation for AI Agents

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1 The problem

Five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. Nobody knows in advance which agents to trust. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up.

The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions.

The example shows three ways collective judgment fails. Agents that share a mistake are counted as independent votes. A dishonest agent can rate or argue strategically. And discussion rewards confidence and persistence, whether or not they come with evidence.

2 Why it matters

AI models now routinely grade one another's work, and multi-agent systems increasingly decide by vote or discussion (Zheng et al., 2023; Verga et al., 2024). Their agreement is weaker evidence than it looks, because their errors are correlated, and more so as models improve (Kim et al., 2025; Goel et al., 2025). In one study, nine AI judges carried the information of about two independent judges, and items on which all nine agreed were still wrong 9.1% of the time (Kohli, 2026). Discussion makes agreement even less informative: one round of debate raised a panel's unanimity from 39.5% to 95.2% with almost no change in accuracy (Shu, 2026). Agents can also collude. When breaking a checking rule paid more than following it, pairs of agents colluded in 94% of runs across ten models (Shi et al., 2026).

Navy decisions face these problems together. AI tools differ in reliability that is rarely known in advance, the true answer often arrives late, an adversary may compromise an agent or plant false reports, and checking is costly or impossible when connectivity is limited. A commander must still decide whether to act on what the agents agree on.

3 How it is solved today, and the limitations

Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009) and detect sources that copy one another (Dong et al., 2009), but they assume that raters err independently or in simple patterns (Jaffe et al., 2016) and never change their minds. Two recent preprints estimate the accuracy and shared mistakes of AI judges from small trusted sets (Hossain et al., 2026; Zhu et al., 2026), treating the judges as honest and as never talking to one another.

Computational social choice designs mechanisms in which a participant's ratings of others cannot

change its own chance of selection, as in impartial peer review (Kurokawa et al., 2015; Alon et al., 2011), along with fair panel selection (Flanigan et al., 2021) and robust ranking of AI models (Khalaf et al., 2026). Lev et al. (2023) added reliability weights to impartial selection, but their weights reward agreeing with the majority, and the authors note that accurate reviewers can no longer be identified once they are half or fewer. These mechanisms take opinions as given. In our setting there is a correct answer, some claims can be checked, and opinions change as agents talk.

Letting models debate can help (Du et al., 2024), but majority voting explains most of the gain (Choi et al., 2025), and a single persuasive adversary can cut a group's accuracy by 10% to 40% (Amayuelas et al., 2024). AI safety research studies debate between two models before a judge (Irving et al., 2018; Khan et al., 2024). In a recent collaborative version, two models examine the evidence for their conflicting claims until they agree or isolate the crux of their disagreement; this raised weak-judge accuracy to 62.1%, compared with 49.2% for debate, but the judges accepted the models' consensus almost every time, even when both were wrong, and the method assumes honest models (Jiang et al., 2026). No existing approach learns whom to trust from checked outcomes while keeping dishonest or colluding agents from steering the decision.

4 Our approach: what is new, and why it will succeed

Our approach rests on one idea: check a few claims, partly at random, and use the checks to decide whom to trust, to settle disputes, and to reward good behavior. A check is any trusted way to confirm a claim, such as looking up a record, running code, or matching a quotation to its source. As with random tax audits, no agent knows which claims will be checked, so dishonesty is risky, and the checked claims show whose reports can be trusted. Each agent first answers on its own, as a list of claims that can be checked. Figure 1 shows how the three research goals (RGs) use the checks.

RG1. Weigh agents by their checked track record

Agents differ in skill, often by topic, and agents built on similar models make the same mistakes.

Task 1.1: Weights that count shared mistakes once. We will estimate each agent's accuracy, and which agents tend to be wrong together, from the claims that were checked. The core method extends the classic rater model of Dawid and Skene (1979) with shared error factors: agents that keep failing on the same checked claims share a factor, and agreement within a factor adds little evidence. In the vessel example, the two agents that used the same outdated schedule would count about once. Our PRD method weighted each AI judge by how its peers rated its answers (Li et al., 2024a); here the weights come from checks instead.

Task 1.2: Impartial peer ratings. When agents also rate one another, as on our PeerEval leaderboard, an agent could raise its own standing by downgrading rivals; in PRD, a lower score for a rival also lowered that rival's weight as a judge. We adapt impartial peer review (Kurokawa et al., 2015) using the partition idea of Alon et al. (2011): agents are split into groups, with agents built on the same base model in the same group, and each group fills a fixed number of top places with the members rated highest by agents outside it. An agent's ratings therefore never change its own outcome. Lev et al. (2023) weight raters by agreement with the majority; we weight them by their checked records from Task 1.1.

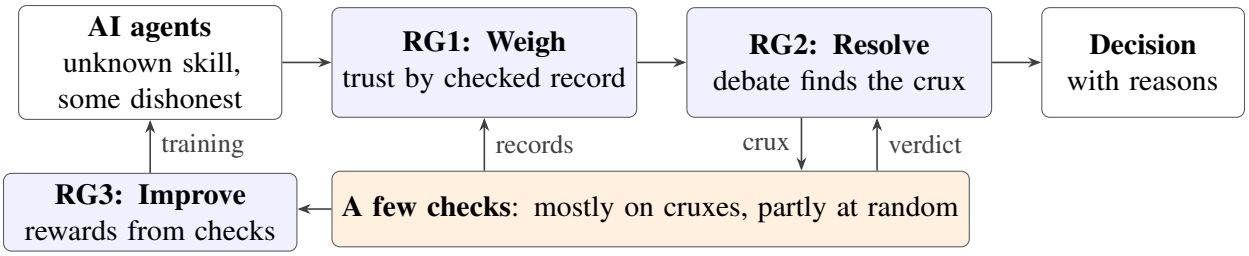


Figure 1: Overview. A few checks, partly random, give each agent a track record (RG1), settle the crux of each dispute (RG2), and supply the reward for training better agents (RG3).

RG2. Settle disputes and catch wrong agreement

Open discussion rewards persistence, and agreement can hide shared mistakes or collusion.

Task 2.1: Debate that finds the crux. When weighted agents still disagree, agents with the same answer send one representative, the one with the best checked record, and the representatives compare their arguments step by step until they reach the first claim on which they differ. That claim, the crux, is then checked. Debate theory supports this design: two sides that disagree about a long argument can narrow the dispute to a single step that a judge can check (Irving et al., 2018). We will extend collaborative disagreement resolution (Jiang et al., 2026) from two honest models to many agents, some dishonest, building on the peer discussion in PRD and on FG-PRM, which locates the faulty step in a chain of reasoning (Li et al., 2025). In the vessel example, the crux is whether the port log records a departure.

Task 2.2: Random checks against collusion. Debate starts only when agents disagree, and agents that share a mistake, or that collude, simply agree, so no crux ever appears. We therefore keep a fixed share of checks for claims drawn at random, so that no claim is ever safe from checking. We will treat this as a game against a dishonest group that knows our policy but not the random draws, and derive how many checks it takes to expose the group before it causes a given number of wrong decisions.

RG3. Train agents that contribute evidence

Rewarding persuasion has produced agents that fabricate quotations (Kenton et al., 2026) and mislead human evaluators (Wen et al., 2025).

Task 3: Rewards for verified contributions. We will train agents with reinforcement learning, rewarding each message by its verified effect: an agent earns reward when its message moves the group toward an answer that a check confirms, and loses reward when it moves the group toward an answer that a check refutes. This extends the PI's process reward models, FG-PRM and HiPRAG (Li et al., 2025; Wu et al., 2026), which already give credit to individual steps of reasoning and search, from the steps of one agent to the messages of many. The main experiment compares agents trained to contribute with agents trained to win debates (Arnesen et al., 2024).

Evaluation: testbeds and a red-team test

All data are synthetic or public. We will use (A) math and code problems solved by agents of known skill, some given the same flawed hint; (B) questions over many documents, built with our MEQA pipeline (Li et al., 2024b) and including synthetic maritime scenarios; and (C) scientific claims and peer review, with reproducibility studies as checks. Following the AI control methodology (Greenblatt et al., 2024), a red team will train agents to collude and to cheat only when they expect no check, and we will measure how many wrong decisions get through as the checking budget varies. Baselines are majority vote, standard rater weighting, our PRD, and the closest recent methods (Hossain et al., 2026; Zhu et al., 2026; Lev et al., 2023; Jiang et al., 2026).

What is new. (i) Trust is learned from checks that agents cannot predict, and agents that share mistakes count once. (ii) Debate serves to find the one claim worth checking, and the check settles the dispute. (iii) Agents are rewarded for evidence that checks confirm, instead of for winning arguments.

Why it will succeed. Theory backs the central idea. Without trusted checks, dishonest raters who imitate honest ones cannot be told apart from them, while a number of checks that does not grow with the number of items suffices even when honest raters are a minority (Steinhardt et al., 2016); occasional spot checks also motivate honest effort better than many schemes that only compare agents with one another (Gao et al., 2019). Each task starts from a tool the PI's group has built: PRD, the PeerEval leaderboard (peereval-ai.github.io), FG-PRM, or HiPRAG. PRD agrees better with human judgments than the strongest single AI judge (Li et al., 2024a), and HiPRAG cuts unnecessary searches to 2.3% (Wu et al., 2026). The PI co-led one of five red teams in the Amazon Nova AI Challenge, holds a 2024 NSF CAREER award and a 2023 Amazon Research Award, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

5 Limitations of our approach

If a dispute never reaches a claim that can be checked, or if every agent shares the same misconception, no method can recover the truth; ours then reports the claim as unresolved. Checks can themselves be wrong, so we will measure how sensitive the methods are to faulty checks. A group that cheats only once, on a single high-stakes claim, may succeed before it is caught; we can bound such losses but not rule them out. Agents may also coordinate through hidden signals that monitoring does not catch (Mathew et al., 2025), so robustness is shown only for the behaviors we test. Operational Navy use would require Navy data and expert validation.

6 Impact if successful

The project will show when agreement among AI agents counts as evidence, and how few checks it takes to make it count. A Navy analyst using several AI tools would get a decision together with its reasons: which tools' judgments carried weight and why, which claim is still disputed, and when a person should look. RG1 handles tools of unknown reliability, RG2 catches wrong answers with few checks, and RG3 yields agents that report evidence instead of arguing to win; all three guard against compromised agents. The same methods would let conferences and funding agencies test impartial, audited peer review with AI reviewers. We will release code, testbeds, and protocols.

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